



Industrial LPDDR4/4X Datasheet

ATL4X2G32M1D-46IT

ATL4X4G32M1D-46IT

200-ball FBGA

Revision 1.0

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Revision History

Version	Date	Editor	Remark
V1.0	2023-4-18	eddy	Basic spec and architecture

Notes : This data sheet contains minimum and maximum limits specified over the power supply and temperature range set forth herein. Although considered final, these specifications are subject to change at any time without notice, as further product development and data characterization sometimes occur.



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1. Product Overview

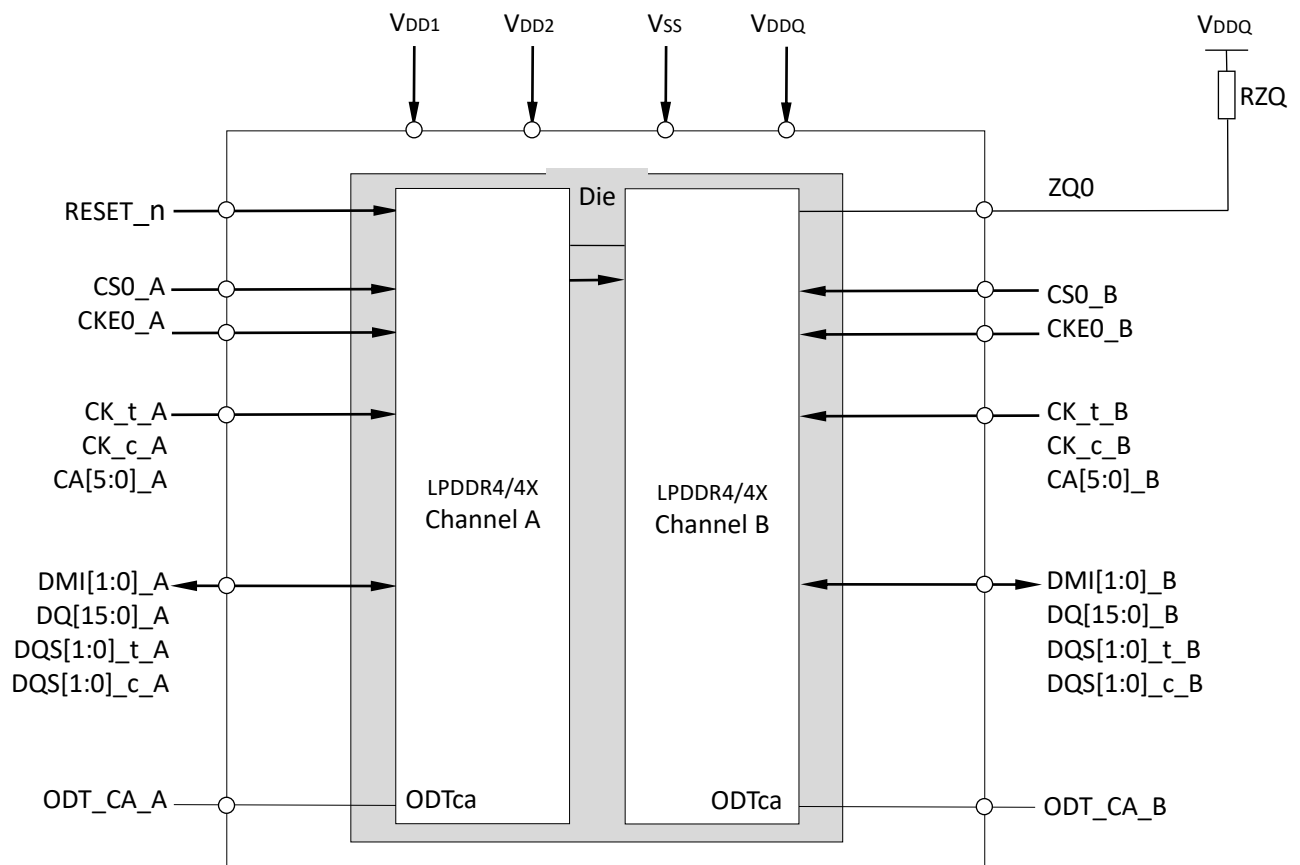
- Ultra-low-voltage core and I/O power supplies
 - $V_{DD1} = 1.70\text{-}1.95\text{V}$; 1.80V nominal
 - $V_{DD2} = 1.06\text{-}1.17\text{V}$; 1.10V nominal
 - $V_{DDQ} = 0.57\text{-}0.65\text{V}$; 0.60V nominal Or $V_{DDQ} = 1.06\text{-}1.17\text{V}$; 1.10V nominal
- Frequency range
 - 2133-10 MHz (data rate range per pin:4266-20Mbps)
- 16n prefetch DDR architecture
- 8 internal banks per channel for concurrent operation
- Single-data-rate CMD / ADR entry
- Bidirectional / differential data strobe per byte lane
- Programmable READ and WRITE latencies (RL / WL)
- Programmable and on-the-fly burst lengths (BL = 16, 32)
- Directed per-bank refresh for concurrent bank operation and ease of command scheduling
- Up to 8.53 GB / s per die x16 channel
- On-chip temperature sensor to control self refresh rate
- Partial-array self refresh (PASR)
- Selectable output drive strength (DS)
- Clock-stop capability
- RoHS-compliant, “green” packaging
- Programmable V_{SS} (ODT) termination
- Single-ended CK and DQS support
- Array configuration
 - 2GB: 512 Meg x 32 (2 channels x 16 I/O)
 - 4GB: 1 Gig x 32 (2 channels x 16 I/O)
- Device configuration
 - 2GB: 512M32 x 1 die in package
 - 4GB: 512M32 x 2 die in package
- “green” package
 - 200-ball FBGA (10mm x 14.5mm x Seated height 1.1mm Max)
- Speed grade,cycle time
 - 468ps@ RL = 36/40

- Temperature range
 - Operating temperature range : -40°C to + 95°C
 - Storage temperature range : -55°C to + 125°C

2. Physical Specifications

2.1. Package Block Diagrams

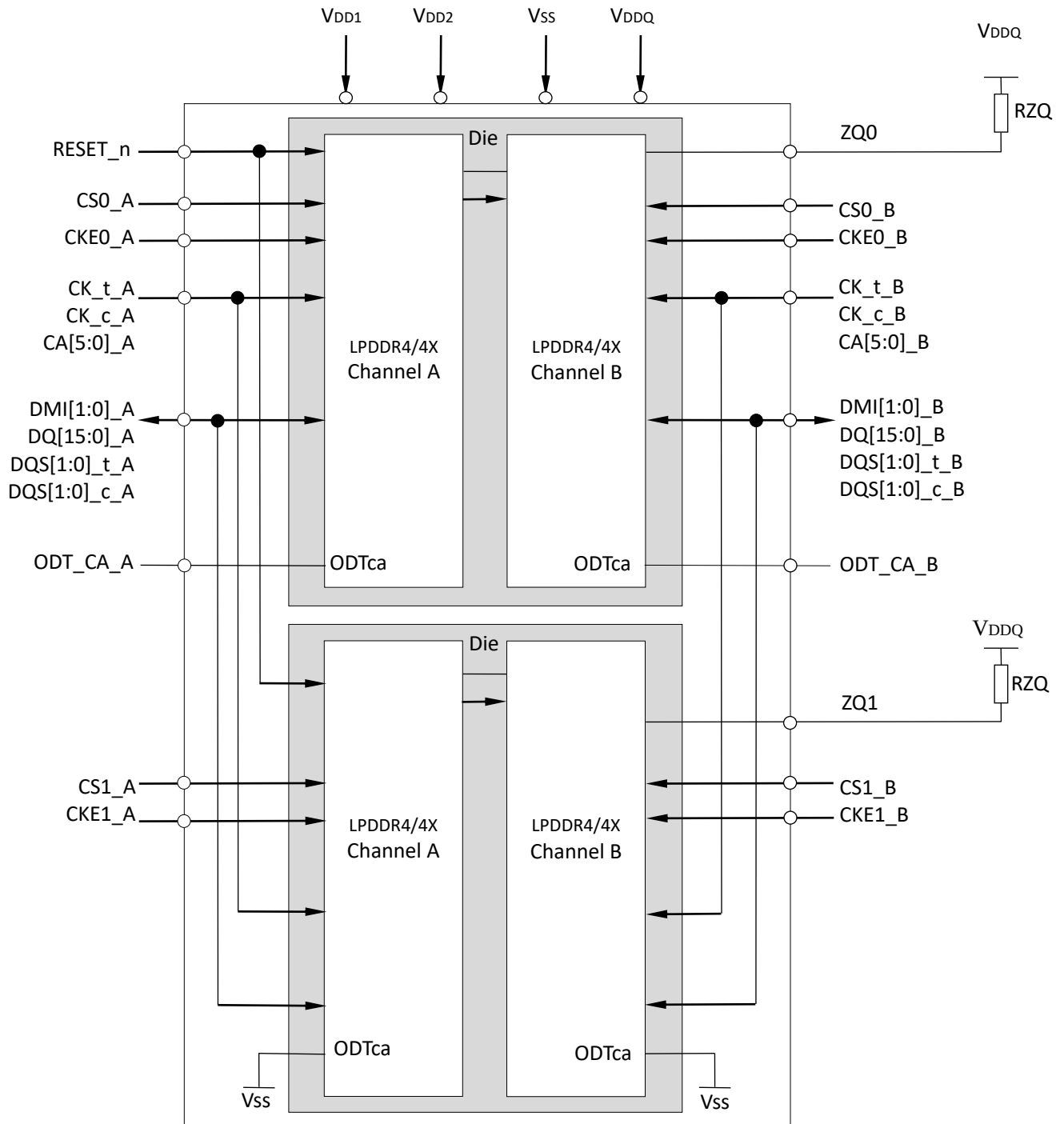
Single-Die, Dual-Channel, Single-Rank Package Block Diagram





Package Block Diagrams (continue)

Dual-Die, Dual-Channel, Dual-Rank Package Block Diagram





2.2. Ball Assignments and Descriptions

200-Ball Single-Die, Dual-Channel, Single-Rank Discrete FBGA

	1	2	3	4	5	6	7	8	9	10	11	12
A	DNU	DNU	V _{SS}	V _{DD2}	ZQ0			NC	VDD2	V _{SS}	DNU	DNU
B	DNU	DQ0_A	V _{DDQ}	DQ7_A	V _{DDQ}			V _{DDQ}	DQ15_A	V _{DDQ}	DQ8_A	DNU
C	V _{SS}	DQ1_A	DMI0_A	DQ6_A	V _{SS}			V _{SS}	DQ14_A	DMI1_A	DQ9_A	V _{SS}
D	V _{DDQ}	V _{SS}	DQS0_t_A	V _{SS}	V _{DDQ}			V _{DDQ}	V _{SS}	DQS1_t_A	V _{SS}	V _{DDQ}
E	V _{SS}	DQ2_A	DQS0_c_A	DQ5_A	V _{SS}			V _{SS}	DQ13_A	DQS1_c_A	DQ10_A	V _{SS}
F	V _{DD1}	DQ3_A	V _{DDQ}	DQ4_A	V _{DD2}			V _{DD2}	DQ12_A	V _{DDQ}	DQ11_A	V _{DD1}
G	V _{SS}	ODT_CA_A	V _{SS}	V _{DD1}	V _{SS}			V _{SS}	V _{DD1}	V _{SS}	NC	V _{SS}
H	V _{DD2}	CA0_A	NC	CS0_A	V _{DD2}			V _{DD2}	CA2_A	CA3_A	CA4_A	V _{DD2}
J	V _{SS}	CA1_A	V _{SS}	CKE0_A	NC			CK_t_A	CK_c_A	V _{SS}	CA5_A	V _{SS}
K	V _{DD2}	V _{SS}	V _{DD2}	V _{SS}	NC			NC	V _{SS}	V _{DD2}	V _{SS}	V _{DD2}
L												
M												
N	V _{DD2}	V _{SS}	V _{DD2}	V _{SS}	NC			NC	V _{SS}	V _{DD2}	V _{SS}	V _{DD2}
P	V _{SS}	CA1_B	V _{SS}	CKE0_B	NC			CK_t_B	CK_c_B	V _{SS}	CA5_B	V _{SS}
R	V _{DD2}	CA0_B	NC	CS0_B	V _{DD2}			V _{DD2}	CA2_B	CA3_B	CA4_B	V _{DD2}
T	V _{SS}	ODT_CA_B	V _{SS}	V _{DD1}	V _{SS}			V _{SS}	V _{DD1}	V _{SS}	RESET_n	V _{SS}
U	V _{DD1}	DQ3_B	V _{DDQ}	DQ4_B	V _{DD2}			V _{DD2}	DQ12_B	V _{DDQ}	DQ11_B	V _{DD1}
V	V _{SS}	DQ2_B	DQS0_c_B	DQ5_B	V _{SS}			V _{SS}	DQ13_B	DQS1_c_B	DQ10_B	V _{SS}
W	V _{DDQ}	V _{SS}	DQS0_t_B	V _{SS}	V _{DDQ}			V _{DDQ}	V _{SS}	DQS1_t_B	V _{SS}	V _{DDQ}
Y	V _{SS}	DQ1_B	DMI0_B	DQ6_B	V _{SS}			V _{SS}	DQ14_B	DMI1_B	DQ9_B	V _{SS}
AA	DNU	DQ0_B	V _{DDQ}	DQ7_B	V _{DDQ}			V _{DDQ}	DQ15_B	V _{DDQ}	DQ8_B	DNU
AB	DNU	DNU	V _{SS}	V _{DD2}	V _{SS}			V _{SS}	V _{DD2}	V _{SS}	DNU	DNU
	1	2	3	4	5	6	7	8	9	10	11	12

Top View (ball down)

 LPDDR4_A(Channel A)  LPDDR4_B(Channel B)  ZQ,ODT_CA,RESET  Supply  Ground



Ball Assignments and Descriptions (continue)

200-Ball Dual-Die, Dual-Channel, Dual-Rank Discrete FBGA

	1	2	3	4	5	6	7	8	9	10	11	12
A	DNU	DNU	V _{SS}	V _{DD2}	ZQ0			ZQ1	V _{DD2}	V _{SS}	DNU	DNU
B	DNU	DQ0_A	V _{DDQ}	DQ7_A	V _{DDQ}			V _{DDQ}	DQ15_A	V _{DDQ}	DQ8_A	DNU
C	V _{SS}	DQ1_A	DMI0_A	DQ6_A	V _{SS}			V _{SS}	DQ14_A	DMI1_A	DQ9_A	V _{SS}
D	V _{DDQ}	V _{SS}	DQS0_t_A	V _{SS}	V _{DDQ}			V _{DDQ}	V _{SS}	DQS1_t_A	V _{SS}	V _{DDQ}
E	V _{SS}	DQ2_A	DQS0_c_A	DQ5_A	V _{SS}			V _{SS}	DQ13_A	DQS1_c_A	DQ10_A	V _{SS}
F	V _{DD1}	DQ3_A	V _{DDQ}	DQ4_A	V _{DD2}			V _{DD2}	DQ12_A	V _{DDQ}	DQ11_A	V _{DD1}
G	V _{SS}	ODT_CA_A	V _{SS}	V _{DD1}	V _{SS}			V _{SS}	V _{DD1}	V _{SS}	NC	V _{SS}
H	V _{DD2}	CA0_A	CS1_A	CS0_A	V _{DD2}			V _{DD2}	CA2_A	CA3_A	CA4_A	V _{DD2}
J	V _{SS}	CA1_A	V _{SS}	CKE0_A	CKE1_A			CK_t_A	CK_c_A	V _{SS}	CA5_A	V _{SS}
K	V _{DD2}	V _{SS}	V _{DD2}	V _{SS}	NC			NC	V _{SS}	V _{DD2}	V _{SS}	V _{DD2}
L												
M												
N	V _{DD2}	V _{SS}	V _{DD2}	V _{SS}	NC			NC	V _{SS}	V _{DD2}	V _{SS}	V _{DD2}
P	V _{SS}	CA1_B	V _{SS}	CKE0_B	CKE1_B			CK_t_B	CK_c_B	V _{SS}	CA5_B	V _{SS}
R	V _{DD2}	CA0_B	CS1_B	CS0_B	V _{DD2}			V _{DD2}	CA2_B	CA3_B	CA4_B	V _{DD2}
T	V _{SS}	ODT_CA_B	V _{SS}	V _{DD1}	V _{SS}			V _{SS}	V _{DD1}	V _{SS}	RESET_n	V _{SS}
U	V _{DD1}	DQ3_B	V _{DDQ}	DQ4_B	V _{DD2}			V _{DD2}	DQ12_B	V _{DDQ}	DQ11_B	V _{DD1}
V	V _{SS}	DQ2_B	DQS0_c_B	DQ5_B	V _{SS}			V _{SS}	DQ13_B	DQS1_c_B	DQ10_B	V _{SS}
W	V _{DDQ}	V _{SS}	DQS0_t_B	V _{SS}	V _{DDQ}			V _{DDQ}	V _{SS}	DQS1_t_B	V _{SS}	V _{DDQ}
Y	V _{SS}	DQ1_B	DMI0_B	DQ6_B	V _{SS}			V _{SS}	DQ14_B	DMI1_B	DQ9_B	V _{SS}
AA	DNU	DQ0_B	V _{DDQ}	DQ7_B	V _{DDQ}			V _{DDQ}	DQ15_B	V _{DDQ}	DQ8_B	DNU
AB	DNU	DNU	V _{SS}	V _{DD2}	V _{SS}			V _{SS}	V _{DD2}	V _{SS}	DNU	DNU
	1	2	3	4	5	6	7	8	9	10	11	12

Top View (ball down)

 LPDDR4_A(Channel A)  LPDDR4_B(Channel B)  ZQ,ODT_CA,RESET  Supply  Ground



2.3. Pad Definition

Symbol	Type	Description
CK_t_A, CK_c_A CK_t_B, CK_c_B	Input	Clock: CK_t and CK_c are differential clock inputs. All address, command and control input signals are sampled on positive edge of CK_t and the negative edge of CK_c.AC timings for CA parameters are referenced to clock. Each channel (A, B) has its own clock pair.
CKE0_A, CK1_A CKE0_B, CK1_B	Input	Clock enable: CKE HIGH activates and CKE LOW deactivates the internal clock signals, input buffers, and output drivers. Power-saving modes are entered and exited via CKE transitions. CKE is sampled at the rising edge of CK.
CS0_A, CS1_A CS0_B, CS1_B	Input	Chip select: Each channel (A, B) has its own CS signals.
CA[5:0]_A CA[5:0]_B	Input	Command/address inputs: Provide the command and address inputs according to the command truth table. Each channel (A, B) has its own CA signals.
ODT_CA_A ODT_CA_B	Input	LPDDR4 CA ODT control: The ODT_CA pin is used in conjunction with the mode register to turn on/off the on-die termination for CA pins. It is bonded to VDD2 within the package, or at the package ball, for the terminating rank, and the non-terminating ranks are bonded to VSS (or left floating with a weak pull-down on the DRAM die). The terminating rank is the DRAM that terminates the CA bus for all die on the same channel. LPDDR4X CA ODT Control: The ODT_CA pin is ignored by LPDDR4X devices. CA ODT is fully controlled through MR11 and MR22. The ODT_CA pin shall be connected to a valid logic level.
DQ[15:0]_A DQ[15:0]_B	I/O	Data input/output: Bidirectional data bus.
DQS[1:0]_t_A DQS[1:0]_c_A DQS[1:0]_t_B DQS[1:0]_c_B	I/O	Data strobe: DQS_t and DQS_c are bidirectional differential output clock signals used to strobe data during a READ or WRITE. The data strobe is generated by the DRAM for a READ and is edge-aligned with data. The data strobe is generated by the SoC memory controller for a WRITE and is trained to precede data. Each byte of data has a data strobe signal pair. Each channel (A, B) has its own DQS_t and DQS_c strobes.
DMI[1:0]_A DMI[1:0]_B	I/O	Data mask/Data bus inversion: DMI is a dual use bidirectional signal used to indicate data to be masked, and data which is inverted on the bus. For data bus inversion(DBI), the DMI signal is driven HIGH when the data on the data bus is inverted, or driven LOW when the data is in its normal state. DBI can be disabled via a mode register setting. For data mask, the DMI signal is used in combination with the data lines to indicate data to be masked in a MASK WRITE command (see the Data Mask (DM) and Data Bus Inversion (DBI) sections for details). The data mask function can be disabled via a mode register setting. Each byte of data has a DMI signal. Each channel has its own DMI signals.



Pad Definition (continue)

Symbol	Type	Description
ZQ0, ZQ1	Reference	ZQ calibration reference: Used to calibrate the output drive strength and the termination resistance. The ZQ pin shall be connected to V_{DDQ} through a $240\Omega \pm 1\%$ resistor.
V_{DDQ} , V_{DD1} , V_{DD2}	Supply	Power supplies: Isolated on the die for improved noise immunity.
V_{SS}	Supply	Ground reference: Power supply ground reference.
RESET_n	Input	RESET: When asserted LOW, the RESET pin resets all channels of the die.
DNU	–	Do not use: Must be grounded or left floating.
NC	–	No connect: Not internally connected.

Notes : LPDDR4X pad definitions are the same as LPDDR4, except ODT_CA pins as described in the Table.

10mm * 14.5mm * 1.10mm Max (Package Code: D)

200X $\varnothing 0.363 \pm 0.05$

$\varnothing 0.15$ (M) A B C
 $\varnothing 0.05$ (M) A

4

12 11 10 9 8 5 4 3 2 1

A B C D E F G H J K L M N P R T U V W Y AA AB

14.5

13.65 CTR

0.65 TYP

0.8 TYP

8.8 CTR

10

B

C

PIN A1 INDEX

0.227 $\pm$ 0.05

(1.10 MAX)

6

3

2

5

5 MAX

4.4 MAX

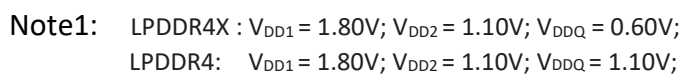
0.9 MAX

1.5 MAX

3.



3.1. Part Number Decoding





3.2. Ordering Options

Table 1: Key Timing Parameters

Speed Grade	Clock Rate (MHz)	Data Rate (Mb/s/pin)	WRITE Latency		READ Latency	
			Set A	Set B	DBI Disabled	DBI Enabled
-46	2133	4266	18	34	36	40

Table 2: Part Number List

Part Number	Total Density	Data Rate	Operating temperature
ATL4X2G32M1D-46IT	2GB(16Gb)	4266 Mb/s/pin	-40°C to + 95°C
ATL4X4G32M1D-46IT	4GB(32Gb)	4266 Mb/s/pin	-40°C to + 95°C

Table 3: Refresh Requirement Parameters

Parameter	Symbol	8Gb Per Channel	unit
REFRESH cycle time (all banks)	t_{RFCab}	280	ns
REFRESH cycle time (per bank)	t_{RFCpb}	140	ns
Per bank refresh to per bank refresh time (different bank)	$t_{PBR2PBR}$	90	ns

Notes : This table only describes refresh parameters which are density dependent. Refer to Refresh Requirement section in General LPDDR4X specification for all the refresh parameters.



3.3. Die Addressing Table

Configuration		2G32 (16Gb/package)	4G32 (32Gb/package)
Die Configuration in the package	Channel A, rank 0	x16 mode × 1 die (dual channel)	x16 mode × 1 die (dual channel)
	Channel B, rank 0		
	Channel A, rank 1	-	x16 mode × 1 die (dual channel)
	Channel B, rank 1		
Die Addressing	Memory density (per die)	16Gb	16Gb
	Memory density (per channel)	8Gb	8Gb
	Configuration	64Mb × 16 DQ × 8 banks × 2channels	64Mb × 16 DQ × 8 banks × 2channels
	Number of channels (per die)	2	2
	Number of banks (per channel)	8	8
	Array prefetch (bits, per channel)	256	256
	Number of rows (per channel)	65,336	65,336
	Number of columns (fetch boundaries)	64	64
	Page size (bytes)	2048	2048
	Channel density (bits per channel)	8,589,934,592	8,589,934,592
	Total density (bits per die)	17,179,869,184	17,179,869,184
	Bank address	BA[2:0]	BA[2:0]
	Row address	R[15:0]	R[15:0]
	Column address	C[9:0]	C[9:0]
	Burst starting address boundary	64-bit	64-bit

Notes : Refer to Package Block Diagrams section in Product specification
and SDRAM Addressing Section in General LPDDR4X specification.



3.4. Mode Register Contents

MR0		
OP7		OP[0] = 1b: Only modified refresh mode supported OP[1] = 0b: Device supports normal latency OP[2] = 0b: Device supports TRR OP[5] = 1b: Device supports single-ended mode
OP6		
OP5	Single-ended mode	
OP4		
OP3		
OP2	RFM support	
OP1	Latency Mode	
OP0	REF	
MR5		
OP7	Manufacturer ID	1111 1111b : Micron
OP6		
OP5		
OP4		
OP3		
OP2		
OP1		
OP0		
MR8		
OP7	I/O width	OP[7:6] = 00b: x16/channel
OP6		
OP5	Density	OP[5:2] = 0100b: 16Gb dual-channel die
OP4		
OP3		
OP2		
OP1		
OP0		

MR3		
OP7		OP[2]= 0b: PPR protection disabled (default) 1b: PPR protection enabled
OP6		
OP5		
OP4		
OP3		
OP2	PPRP ³	
OP1		
OP0		
MR6		
OP7	Revision ID1	0000 0111b
OP6		
OP5		
OP4		
OP3		
OP2		
OP1		
OP0		
MR13		
OP7		OP[2]= 0b: Normal operation (default) 1b: Output the V _{REF(CA)} value on DQ7 and V _{REF(DQ)} value on DQ6
OP6		
OP5		
OP4		
OP3		
OP2	VR0	
OP1		
OP0		



Mode Register Contents (continue)

MR24		
OP7	TRR Mode	OP[3:0] = 1000b: Unlimited MAC OP[7] = 0b: Disable (default) 1b: Reserved
OP6		
OP5		
OP4		
OP3	Unlimited MAC	
OP2	MAC Value	
OP1		
OP0		

MR25			
OP7	PPR resources ⁴	Bank 7	0b: PPR resource is not available 1b: PPR resource is available
OP6		Bank 6	
OP5		Bank 5	
OP4		Bank 4	
OP3		Bank 3	
OP2		Bank 2	
OP1		Bank 1	
OP0		Bank 0	

- Notes:**
1. The contents of Product Specific Mode Register definition will reflect information specific to each die in these package
 2. Other bits not defined above and other mode registers are referred to in Mode Register Assignments and Definitions section.
 3. When not using PPR function, PPR protection should be enabled to prevent unintended PPR entry (MR3 OP[2] = 1b).
 4. Before using PPR function, confirm the availability of PPR resource by reading MR25.